MAE 6246 – Homework 7

Part 1: Point-Mass LQR by Hand

Name: _____ **Part 1:** 40 points

Due: Friday, November 13, 2026, at 10:00 AM (Eastern Time).

Submission. Complete these seven subproblems by hand, show each algebraic step, and submit your clearly handwritten work as one PDF through Blackboard. Part 2 (60 points) is specified in the companion notebook, `homework07_furuta_control.ipynb`. Submit the completed notebook alongside your handwritten Part 1 PDF. The full assignment is worth 100 points.

Late policy. There is a penalty of **1 point per 10 minutes late**, enforced strictly except under documented extraordinary circumstances. The course policy on generative AI applies; submit the required supplementary record when AI materially influences your work.

Part 1 – Point-mass LQR by hand (40 points)

Consider a unit point mass with position x and force input u:

$$\ddot{x} = u, \qquad z = \begin{bmatrix} x \\ v \end{bmatrix}, \qquad v = \dot{x}.$$

We seek an infinite-horizon regulator minimizing

$$J = \frac{1}{2}\int_0^\infty \left(q_x x^2 + q_v v^2 + r u^2\right) dt, \qquad q_x > 0, \quad q_v \geq 0, \quad r > 0.$$

The following subproblems guide a two-state algebraic Riccati calculation. Use the continuous-time equation and feedback convention

$$A^T P + PA - PBR^{-1}B^T P + Q = 0, \qquad u = -Kz, \qquad K = R^{-1}B^T P.$$

(a) **State-space model (4 points).** Write $\dot{z} = Az + Bu$, specifying the dimensions and entries of A and B.

(b) **Quadratic objective (4 points).** Express J as $\frac{1}{2}\int_0^\infty (z^T Q z + u^T R u) dt$ and identify Q and R. State whether each matrix is positive definite or positive semidefinite under the given assumptions, including the case $q_v = 0$.

(c) **Introduce the unknown matrix (8 points).** Let

$$P = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_3 \end{bmatrix}.$$

Compute $A^T P$, PA, and $PBR^{-1}B^T P$ separately. Show the matrix multiplications before substituting them into the Riccati equation.

Part 1 – continued

(d) **Three scalar equations (6 points).** Equate the $(1,1)$, $(1,2)$, and $(2,2)$ entries of the Riccati equation to zero. Explain why the $(2,1)$ entry gives no additional independent equation. First solve for the possible values of p_2. Then express p_3^2 in terms of p_2, and p_1 in terms of p_2, p_3.

(e) **Select and verify the positive-definite solution (8 points).** Choose the signs of p_2 and p_3 so that $P \succ 0$ and obtain explicit formulas for p_1, p_2, p_3 in terms of q_x, q_v, r. Do not choose a square-root branch without justification.

For a real symmetric 2×2 matrix,

$$P \succ 0 \quad \Longleftrightarrow \quad p_1 > 0 \text{ and } p_1 p_3 - p_2^2 > 0.$$

A positive-definite matrix also has $p_3 > 0$. Use these facts with your scalar equations to eliminate inadmissible sign choices, and verify both inequalities for your final expression. Your argument must also cover $q_v = 0$.

(f) **Feedback and a numerical example (6 points).** Find K symbolically. Set $q_x = q_v = r = 1$ and compute P, K, the characteristic polynomial of $A - BK$, and its two eigenvalues by hand. Verify that the closed-loop matrix is Hurwitz.

(g) **Interpret the optimal cost (4 points).** Use $J^* = \frac{1}{2} z_0^T P z_0$ to calculate the minimum cost for the numerical example with $z_0 = [1, 0]^T$. How does this cost change if the initial displacement doubles while the initial velocity remains zero? Explain how multiplying Q and R by the same positive constant affects P, K, and the optimal trajectory.